MTH1001 Mathematical Structures 2025–26, Term 1

Problem Sheet 3

This sheet is for discussion in the week 8 tutorials (Wednesday 12th November). Be ready to discuss (at least) Questions A4, A7(c), A8(b), B1, B5(a)(c) in the tutorial.

Your answers should use mathematical notation correctly and be written in logically correct sentences, including words such as "let", "if", "then". You should introduce any variables that occur in your argument.

SECTION A

These are fairly routine questions. There will be some questions on the January exam of a similar difficulty to these, and some more challenging questions.

A1. The following table defines three permutations σ, τ, ρ in S_8:

j	1	2	3	4	5	6	7	8
$\sigma(j)$	2	3	1	5	6	7	4	8
$\tau(j)$	4	2	3	8	5	6	7	1
$\rho(j)$	4	2	3	1	7	6	5	8

(a) Write σ, τ and ρ as products of disjoint cycles.

(b) Calculate $\sigma\tau$, $\tau\sigma$, $\rho\sigma$, $\sigma\rho\tau$ by multiplying out the cycles, giving your answers as products of disjoint cycles.

(c) Write the inverse of each of the permutations σ, τ, ρ, $\sigma\tau$, $\rho\sigma$, $\sigma\rho\tau$ as a product of disjoint cycles.

(d) Find the order and the sign of each of the permutations σ, τ, ρ, $\sigma\tau$, $\rho\sigma$, $\sigma\rho\tau$.

A2. How many elements does the set S_4 of all permutations of $\{1, 2, 3, 4\}$ have? Write them all down in cycle notation, and determine the order and sign of each. (You might find it convenient to collect together all those whose decomposition into cycles has the same shape.)

A3. Let $\rho = (15478)(26)$ and $\sigma = (1357)(2468)$ be elements of S_8. Find $\rho\sigma$, $\sigma\rho$, ρ^6 and $\sigma\rho\sigma^{-1}$, and determine the order and sign of each of these permutations.

A4. For $\sigma = (1246)(375)$ and $\tau = (135)(46)$, calculate each of the permutations $\sigma\tau$, $\tau\sigma$, σ^3, τ^{-1}, $\tau\sigma\tau^{-1}$, giving your answers as products of disjoint cycles. Also, find the order and the sign of each of these permutations.

A5. Show that S_{10} contains a permutation of order 21 but no permutation of order 24. Does it contain a permutation of order 18?

A6. Express the following permutations as products of transpositions:

(a) $(13624)(5798)$;

(b) $(124)(365)(789)$;

(c) $(1258439)(67)$.

A7. Use induction to prove that the following statements are true for every natural number n:

(a) $1^3 + 2^3 + 3^3 + \ldots + n^3 = \frac{1}{4}n^2(n+1)^2$;

(b) $n^3 + 5n$ is divisible by 3

(c) $6^n + 20n - 1$ is divisible by 25;

(d) $2^{n+1} + 3^{2n-1}$ is divisible by 7.

A8. Use induction to prove the following summation formulas:

(a) $\displaystyle\sum_{j=1}^{n}(3j-1) = \frac{1}{2}n(3n+1)$;

(b) $\displaystyle\sum_{j=1}^{n}\frac{1}{j(j+1)} = \frac{n}{n+1}$;

(c) $\displaystyle\sum_{j=1}^{n}3^{j-1}j = \frac{1}{4}\left[3^n(2n-1)+1\right]$;

(d) $\displaystyle\sum_{j=1}^{n}j \cdot j! = (n+1)! - 1$.

A9. A sequence of numbers $a_1, a_2, a_3, \ldots$ is defined by $a_1 = 1$, $a_2 = 5$ and

$$a_n = a_{n-1} + 2a_{n-2} \quad \text{for } n \geq 3.$$

Show by (strong) induction that $a_n = (-1)^n + 2^n$.

SECTION B

These are more challenging questions. There may be questions on the January exam of similar difficulty to these.

B1. Prove that a permutation of odd order is necessarily an even permutation. Is a permutation of even order necessarily an odd permutation?

B2. The point of this question is to show that the sign of a permutation is well-defined, i.e. that a permutation $\sigma \in S_n$ cannot be written both as the product of an odd number of transpositions and as the product of an even number of transpositions.

(a) Let $\sigma = (1526)(384)$ be an element of S_8. Let $\tau_1 = (25)$ and $\tau_2 = (78)$ be transpositions in S_8. Calculate $\sigma\tau_1$ and $\sigma\tau_2$ and count how many cycles (including any cycles of length 1) each of these permutations have.

(b) In general, prove that if $\sigma \in S_n$ is a permutation with k cycles in its cycle notation, and τ is a transposition in S_n then $\sigma\tau$ either has $k+1$ cycles or has $k-1$ cycles. [In this question we **must** count cycles of length 1, so the identity permutation in S_n will have n cycles.]

(c) Show that if σ is the product of k disjoint cycles (including any cycles of length 1):

$$\sigma = \sigma_1 \cdots \sigma_k,$$

and if σ can also be written as the product of r transpositions:

$$\sigma = \tau_1 \cdots \tau_r,$$

then $(-1)^{k+r} = (-1)^n$. (*Hint:* First show that $\sigma_1 \cdots \sigma_k \tau_r \cdots \tau_1$ is the identity permutation.)

(d) Deduce that a permutation cannot both be a product of an even number of transpositions and of an odd number of transpositions.

B3. (a) Using only (i) the product rule for differentiation, and (ii) the fact that $\dfrac{d}{dx}(x) = 1$, prove by induction on n that $\frac{d}{dx}(x^n) = nx^{n-1}$ for all $n \in \mathbb{N}$.

(b) Let a be a constant. Assuming that $\frac{d}{dx}(\exp(x)) = \exp(x)$, show by induction that $\frac{d^n}{dx^n}(\exp(ax)) = a^n \exp(ax)$ for all $n \geq 1$.

B4. Find the smallest natural number m such that $m! > 3^m$, and prove by induction that $n! > 3^n$ for all $n \geq m$.

B5. The sequence F_1, F_2, F_3, $\ldots$ of Fibonacci numbers is defined by

$$F_1 = 1, \quad F_2 = 1, \quad F_{k+1} = F_k + F_{k-1} \text{ for } k \geq 2.$$

(a) Write out F_n for $1 \leq n \leq 10$.

(b) Prove that F_{3n} is even for all natural numbers n.

(c) Conjecture and prove a theorem concerning the sum $F_1 + F_3 + F_5 + \cdots + F_{2n-3} + F_{2n-1}$.

(d) Conjecture and prove a theorem concerning the sum $F_2 + F_4 + F_6 \cdots + F_{2n-2} + F_{2n}$.

(e) Prove that

$$F_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^n$$

for all natural numbers n.